Dr. A. P. J. Abdul Kalam Technical University, Uttar Pradesh, Lucknow

POSTGRADUATE COURSE
MASTER OF DESIGN

BATCH 2018-2019

Dr. A. P. J. Abdul Kalam Technical University, Uttar Pradesh, Lucknow

Semester I

	Credits
BDS 511 Design Fundamentals	3
BDS 512 Representation Techniques	4
BDS 513 Product Design I	4
BDS 514 Ergonomics	3
BDS 551 Design Workshop	4
	Credits 18

Semester II

BDS 521 Sustainable Design	4
BDS 522 Materials and Processes in Design	3
BDS 523 Product Design II	4
BDS 524 Design Research Methodology	3
BDS 541 Elective I	4
	Credits 18

Semester III

BDS 611 Design Management	2
BDS 631 Elective II	2
BDS 632 Elective III	2
BDS 633 Elective IV	2
BDS 671 Summer Internship	2
BDS 681 Thesis	8
	Credits 18

Semester IV

BDS 699	Thesis	18
		Credits 18

Total credits 72

Overview

Semester I						
Course Code	**Course Name**	**Course content**	**L**	**T**	**P**	**C**
BDS 511	Design Fundamentals	Elements of Design, Principle of Design, Color and Composition, lights and shades.	1	0	4	3
BDS 512	Representation techniques	Freehand sketching, illustration, Vector and raster Software's	1	0	6	4
BDS 513	Product Design I	Introduction to product development process, Concept generation, concept selection etc.	1	0	12	4
BDS 514	Ergonomics	Physical and cognitive ergonomics	2	0	2	3
BDS 551	Design Workshop	Fitting and carpentry, welding etc.	0	0	8	4
						Credits 18

Semester II						
Course Code	**Course Name**	**Course content**	**L**	**T**	**P**	**C**
BDS 521	Sustainable Design	Sustainable design principles. Physical, mental, spiritual, cultural, social, ethical and economic issues in designing for sustainability. Ecological footprints, ecosystem impact. Waste, reuse and recycling, benign emissions, green design.	1	0	6	4
BDS 522	Material and Processes in Design	Introduction to materials, properties of materials, Ferrous and non-ferrous metals, wood, structure and properties of wood, Plastics: Finishing metals, etc. Molding process, fastening and joining, Composites, Design in Plastics, Casting, Bending, Drawing process, Forging, Machining.	1	0	4	3
BDS 523	Product Design II	Problem solving using a System-based approach – Using knowledge and creativity to develop Product service system (PSS) based solutions to problems that we're currently facing in our society.	1	0	12	4
BDS 524	Design Research Methodology	Qualitative and qualitative research methodology, Questionnaire design, validation, repeatability testing, psychophysical scales, Direct observation and activity analysis, photography as a tool in design research etc.	2	0	2	3
BDS 541	Elective I		1	0	6	4
						Credits 18

Semester III						
Course Code	**Course Name**	**Course content**	**L**	**T**	**P**	**C**
BDS 611	Design Management	Introducing strategic design: Tangible Vs Intangible design, Design thinking vs Strategic thinking, Tools for strategic design, innovation through strategic design, Design process management etc.	1	0	2	2
BDS 631	Elective II		1	0	2	2
BDS 632	Elective III		1	0	2	2
BDS 633	Elective IV		1	0	2	2
BDS 671	Summer Internship		0	0	0	2
BDS 681	Thesis		0	0	16	8
						Credits 18

Semester IV						
Course Code	**Course Name**	**Course content**	**L**	**T**	**P**	**C**
BDS 699	Thesis		0	0	36	18
						Credits 18

SEMESTER I

BDS 511 Design Fundamentals

Elements of Design- point, line, shape, space, form, color, texture etc. Principles of Design- balance, scale, proportion, contrast, emphasis, pattern, movement, rhythm, harmony, and unity on the basic of element of design, Development of patterns by repetition of points, dots, rectilinear elements, curvilinear elements, shapes. Regular and irregular patterns. Development of design aptitude thorough understanding of the elements and principles of design, Shapes- concept of positive and negative shapes. Understanding flat surfaces, colors, pigments and light, additive and subtractive models. Shades of grey, Color dimensions, understanding warm and cool colors, Elements of form, order and structure, development of surfaces, Nature form and bionic, 4D forms, Exposure to various elements of 3D and principles of art and design in 3D, Expression and exploration of space and volumes, exposure to form and movement, Identify and form: creating form family, Material exploration.

Reference books:

Bervin, M. E. (1984). Design Through Discovery: The Element and Principles. Holt, Rinehart and Winston, Washington.

Wong, W. (1972). Principles of two-dimensional design. John Wiley & Sons. Hannah, G. G. (2002). Elements of design: Rowena Reed Kostellow and the structure of visual relationships. Princeton Architectural Press.

Gamma, E. (1995). Design patterns: elements of reusable object-oriented software.

Pearson Gyorgy. K (1995), Language of vision

Kimberly. E (2001), Geometry of Design: Studies on proportions and compositions

BDS 512 Representation Techniques

Different ways in design ideas can be represented for better visualization. Development of an analytical ability to deal with complexity of imagination and visualization of object from any angle – Sketching for ideation, concepts, object representation, figure drawing. One point, two point, and three-point perspective. Understanding and representing the structure of forms in detail with wireframes. Color representation in the object drawing with section and exploded view.

Reference books

Betty .E (2002), New Drawing on the Right Side of the Brain

Wood, P., & McDonnell, P. (1994). Scientific illustration: a guide to biological, zoological, and medical rendering techniques, design, printing, and display. John Wiley & Sons.

Buxton, B. (2010). Sketching user experiences: getting the design right and the right design: getting the design right and the right design. Morgan Kaufmann.

Powell, D. (1990). Presentation techniques. New York: Little, Brown & Company.

Tal, D. (2010). Google Sketch up for site design: a guide to modelling site plans, terrain and architecture. John Wiley & Sons.

Zeman, N. B. (2014). Essential Skills for 3D Modelling, Rendering, and Animation. CRC Press.

BDS 513 Product Design I

Generic product design and development process. Levels of product, Product life cycle, Design and development of a product, by understanding the market, the client, the technology. Product mix and product portfolio, Concept generation, selection and testing, embodiment design, detailed design and prototyping. Developments of products evaluate and refine the prototypes.

Reference Books

Cross, N. (2008). Engineering design methods: strategies for product design. John Wiley & Sons. Roozenburg, N. F., & Eekels, J. (1995). Product design: fundamentals and methods (Vol. 2). Chichester: Wiley.

Cuffaro, D., & Zaksenberg, I. (2013). The Industrial Design Reference & Specification Book:
Everything Industrial Designers Need to Know Every Day. Rockport Publishers.

Mohr, J. J., Sengupta, S., & Slater, S. F. (2009). Marketing of high-technology products and Innovations. Pearson Prentice Hall.

BDS 514 Ergonomics

Definition of Ergonomics and its application and overview, Concept of Man Machine Environment System. Overview of Human body and its sub systems, understanding musculoskeletal system and its function in terms of manual activities, Understanding nervous system, human sensory organs and their limitations, Cognition, perception and performance, Task Analysis, Anthropometry and application in product and space.

Reference Books

Lev Manovich, The Language of New Media, Cambridge, MIT Press, 2001

Joe Lambert, Digital Storytelling: Capturing Lives, Creating Community, Life on the Water Inc, 2008

Noah Wardrip-Fruin (Editor) and Nick Montfort, The New Media Reader, MIT Press, 2003 ISBN 0262232278

Bryan Alexander, The New Digital Storytelling: Creating Narratives with New Media, Praeger, 2011

Robert Klanten, Interactive Installations and Experiences, Die Gestalten Verlag, 2011

B. Peacock and W. Karwowski, Automotive ergonomics, Taylor & Francis Ltd., 1993

William H. Cushman and Daniel J. Rosenberg, Human Factors in Product Design, Elsevier Science Pub Co, 1991

BDS 551 Design Workshop

Emphasis on the skill of workshop methods. Hands on techniques to manipulate the basic materials according their understanding. Exposure to the students on some of the software on Product Design and Visual Design. The focus of the course is teaching the students about design intent and how software can be utilized for the maximum benefit of the designer. The student project involves making products out of concepts in virtual environment.

Reference Books

Williams, R., & Sheldon, C. (2009). Robin Williams Handmade Design Workshop: Create Handmade Elements for Digital Design. Peachpit Press.

Cuffaro, D., & Zaksenberg, I. (2013). The Industrial Design Reference & Specification Book: Everything Industrial Designers Need to Know Every Day. Rockport Publishers.

Sherwin, D. (2010). Creative workshop: 80 challenges to sharpen your design skills. How Books.

Fullerton, T., Swain, C., & Hoffman, S. (2004). Game design workshop: Designing, prototyping, & playtesting games. CRC Press.

SEMESTER II

BDS 521 Sustainable Design

Sustainable design principles. Physical, mental, spiritual, cultural, social, ethical and economic issues in designing for sustainability.

Ecological footprints, ecosystem impact.

Cradle-to cradle, Cradle to grave, Frugal Design through sustainability

Integrated DFE (Design for Environment) /Eco design, design for sustainability, eco innovation, system-wide product/service strategies,

Reference Books

Mendler, S., & Odell, W. (2000). The HOK guidebook to sustainable design. John Wiley & Sons. 4.

Williams, D. E. (2007). Sustainable design: Ecology, architecture, and planning. John Wiley & Sons.

Walker, S. (2012). Sustainable by design: Explorations in theory and practice. Routledge.

Fairs, M. (2009). Green design: creative sustainable designs for the twenty-first century. North Atlantic Books.

BDS 522 Material and Processes in Design

Introduction to materials, properties of materials, Ferrous and non-ferrous metals, wood, structure and properties of wood, Plastics: Thermoplastics and Thermosetting plastics, Classification of thermoplastics and thermosetting plastics, finishing metals, some popular finishes (oil, paint, etc.), Industrial finishes for plastic, wood and metals, Finishing Wood, Composites, and its properties, Influence of materials and processes on product aesthetics.

Moulding process, fastening and joining, Composites, Design in Plastics, Casting, Bending, Drawing process, Forging, Machining. Form exploration with different materials.

Reference Books

Ashby M., Johnson K., 2002, Materials and Design: The Art and Science of Material Selection in Product Design: Butterworth-Heinemann.

Degarmo E. P., Materials and Processes in Manufacturing, 9th ed., John Wiley &Sons, 2002

Beck R. D.: Plastic Product Design, Van Nostrand Reinhold Co., New York, 1980

Garratt J.: Design and Technology, Cambridge University Press, UK, 20004

Thompson R.: Manufacturing processes for design professionals, Thames & Hudson, London 2007

BDS 523 Product Design II

Design solution to problems related to the individual and social application of the product. Product form, production variables and technological process along with step by step analysis of the product design and development process.

Problem solving using a System-based approach – Using knowledge and creativity to develop Product service system (PSS) based solutions to problems. User Centric Design Process, Introduction to Universal Design and its principles.

Reference Books

Hann, M. (2013). Structure and Form in Design: Critical Ideas for Creative Practice. A&C Black.

Boden, M. A. (2012). Creativity and art: three roads to surprise.

Bentley, T. (2012). Learning beyond the classroom: Education for a changing world. Routledge.

Roozenburg, N. F., & Eekels, J. (1995). Product design: fundamentals and methods (Vol. 2). Chichester: Wiley.

BDS 524 Design Research Methodology

Qualitative and qualitative research methodology,

Questionnaire design, validation, repeatability testing, psychophysical scales,

Direct observation and activity analysis, photography as a tool in design research etc.

Reference Books

Laurel, B. (2003). Design research: Methods and perspectives. MIT press.

Koskinen, I., Zimmerman, J., Binder, T., Redstrom, J., &Wensveen, S. (2011). Design research through practice: From the lab, field, and showroom. Elsevier.

Creswell, J. W., & Clark, V. L. P. (2007). Designing and conducting mixed methods research.

Creswell, J. W. (2013). Research design: Qualitative, quantitative, and mixed methods approaches. Sage publications.

SEMESTER III

BDS 611 Design Management

Identification of user needs and Driving Factors, Emotional Design, Sensibility, Social Ethics and Concerns, Market Gaps, Product Planning for the future, Product Life Cycle management, Planning for redesign, Strategy for product differentiation and identity, Intellectual Property Rights (IPR), Briefs, Detailed Briefs and Concept Notes, Cost Estimation and Time management, Documentation and Reporting, Work distribution, Project Management tools

Reference Books

Kathryn Best, The Fundamentals of Design Management, AVA Publishing, 2010

Brigitte Borja De Mozota, Design Management: Using Design to Build Brand Value and Corporate Innovation, Allworth Press, 2004

Kenneth B Khan, Product Planning Essentials, M E Sharpe Inc, 2011

John Stark, Product Lifecycle Management: 21st Century Paradigm for Product Realisation, Springer, 2011

Craig M. Vogel and Jonathan Cagan, Creating Breakthrough Products: Innovation from Product Planning Program Approval, FT Press, 2001

David L. Rainey, Product Innovation: Leading Change through Integrated Product Development, Cambridge University Press, 2011

Best, K. (2006). Design management: managing design strategy, process and implementation. AVA publishing.

Cooper, R., Junginger, S., & Lockwood, T. (Eds.). (2013). The handbook of design management. A&C Black.

Martin, R. L. (2009). The design of business: Why design thinking is the next competitive advantage. Harvard Business Press.

Electives

BDS 541 Elective I
BDS 541a Applied Ergonomics
BDS 541b Aerospace Design
BDS 541c Medical Equipment Design

BDS 631 Elective II
BDS 631 MOOC Course

BDS 632 Elective III
BDS 632a Visual Communication
BDS 632b Typography
BDS 632c Trend research and Forecasting

BDS 633 Elective IV
BDS 633a Interaction Design
BDS 633b Information Design
BDS 633c Packaging Design

Others Electives:
Toy Design
Universal Design
Architectural Studies in Design
Bio Inspired Design
Furniture Design etc.
Art
Psychology
Creativity
Color

EVALUATION SCHEME

SEMESTER I

Sl. No.	Course Code	Subject	Periods			Evaluation Scheme			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	BDS 511	Design Fundamentals	1	0	4	30	70	100	0	100	200	**3**
2	BDS 512	Representation Techniques	1	0	6	30	70	100	0	100	200	**4**
3	BDS 513	Product Design I	1	0	12	30	70	100	0	100	200	**4**
4	BDS 514	Ergonomics	2	0	2	30	70	100	0	100	200	**3**
5	BDS 551	Design Workshop	0	0	8	30	70	100	0	100	200	**4**
		Total	**5**	**0**	**32**						**1000**	**18**

SEMESTER II

Sl. No.	Course Code	Subject	Periods			Evaluation Scheme			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	BDS 521	Sustainable Design	1	0	6	30	70	100	0	100	200	**4**
2	BDS 522	Material and Processes in Design	1	0	4	30	70	100	0	100	200	**3**
3	BDS 523	Product Design II	1	0	12	30	70	100	0	100	200	**4**
4	BDS 524	Design Research and Methodology	2	0	2	30	70	100	0	100	200	**3**
5	BDS 541	Elective I	1	0	6	30	70	100	0	100	200	**4**
		Total	**6**	**0**	**30**						**1000**	**18**

SEMESTER III

Sl. No.	Course Code	Subject	Periods			Evaluation Scheme			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	BDS 611	Design Management	1	0	4	30	70	100	0	50	150	**2**
2	BDS 631	Elective II	1	0	4	30	70	100	0	50	150	**2**
3	BDS 632	Elective III	1	0	2	30	70	100	0	50	150	**2**
4	BDS 633	Elective IV	1	0	2	30	70	100	0	50	150	**2**
5	BDS 671	Summer Internship	0	0	0	0	0	0	0	100	100	**2**
6	BDS 681	Thesis	0	0	16	0	100	100	0	200	300	**8**
		Total	**4**	**0**	**28**						1000	**18**

SEMESTER IV

Sl. No.	Course Code	Subject	Periods			Evaluation Scheme			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	BDS 699	Design Management	0	0	36	0	500	500	0	500	1000	**18**
		Total	**0**	**0**	**36**						1000	**18**